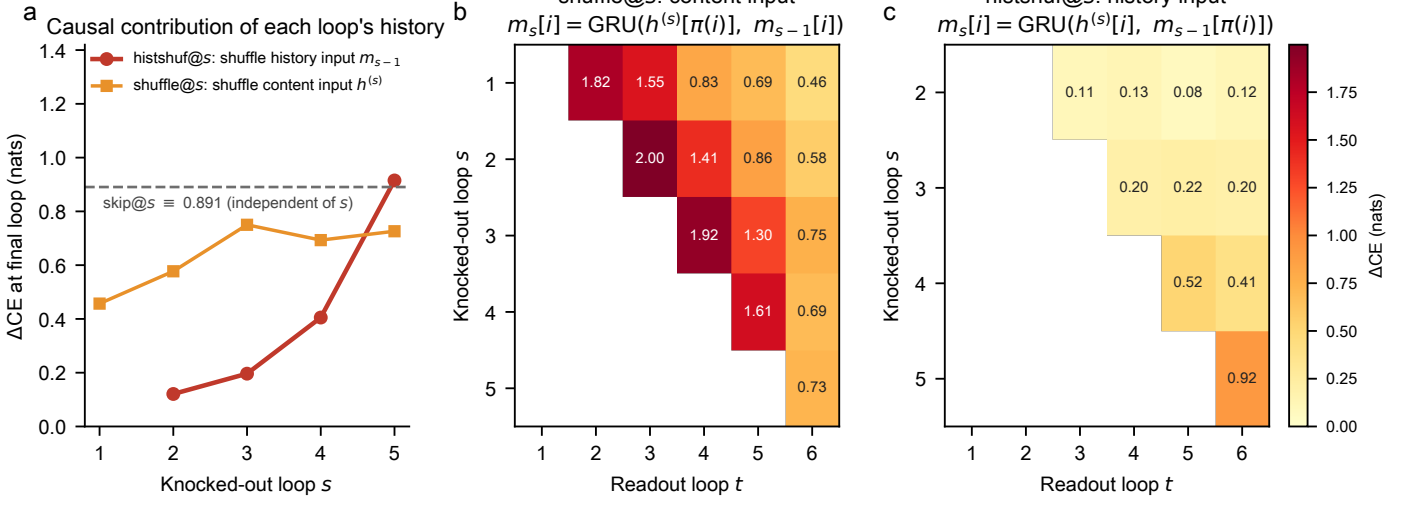
a
Causal contribution of each loop's history
histshuf@s: shuffle history input m_{s-1}
shuffle@s: shuffle content input $h^{(s)}$
skip@s ≡ 0.891 (independent of s)
ΔCE at final loop (nats)
Knocked-out loop s
b
shuffle@s: content input
$m_s[i] = \mathrm{GRU}(h^{(s)}[\pi(i)],\ m_{s-1}[i])$
1.82 1.55 0.83 0.69 0.46
2.00 1.41 0.86 0.58
1.92 1.30 0.75
1.61 0.69
0.73
Knocked-out loop s
Readout loop t
c
histshuf@s: history input
$m_s[i] = \mathrm{GRU}(h^{(s)}[i],\ m_{s-1}[\pi(i)])$
0.11 0.13 0.08 0.12
0.20 0.22 0.20
0.52 0.41
0.92
Knocked-out loop s
Readout loop t
ΔCE (nats)